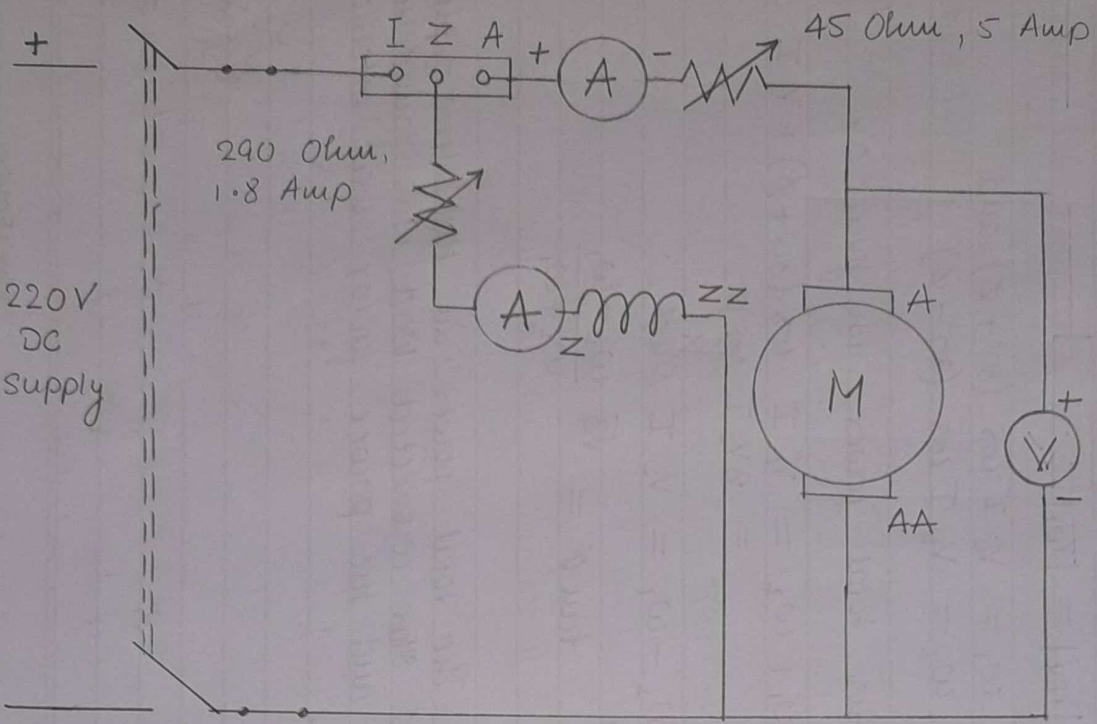


AIM: To study speed control methods of DC shunt motor.

CIRCUIT DIAGRAM:



Exp No. 11

Date 10/3/22
Page No. 33
Signature

AIM: To study speed control methods of DC shunt motor.

APPARATUS:

Motor : DC shunt motor, 2HP, 280, 8A, 1500 RPM
Ammeter (MC) : 0-2 A
Voltmeter (MC) : 0-300 V
Rheostat : 240 Ohms, 1.8 A (No. 1)
: 45 Ohms, 5 A (No. 1)

THEORY:

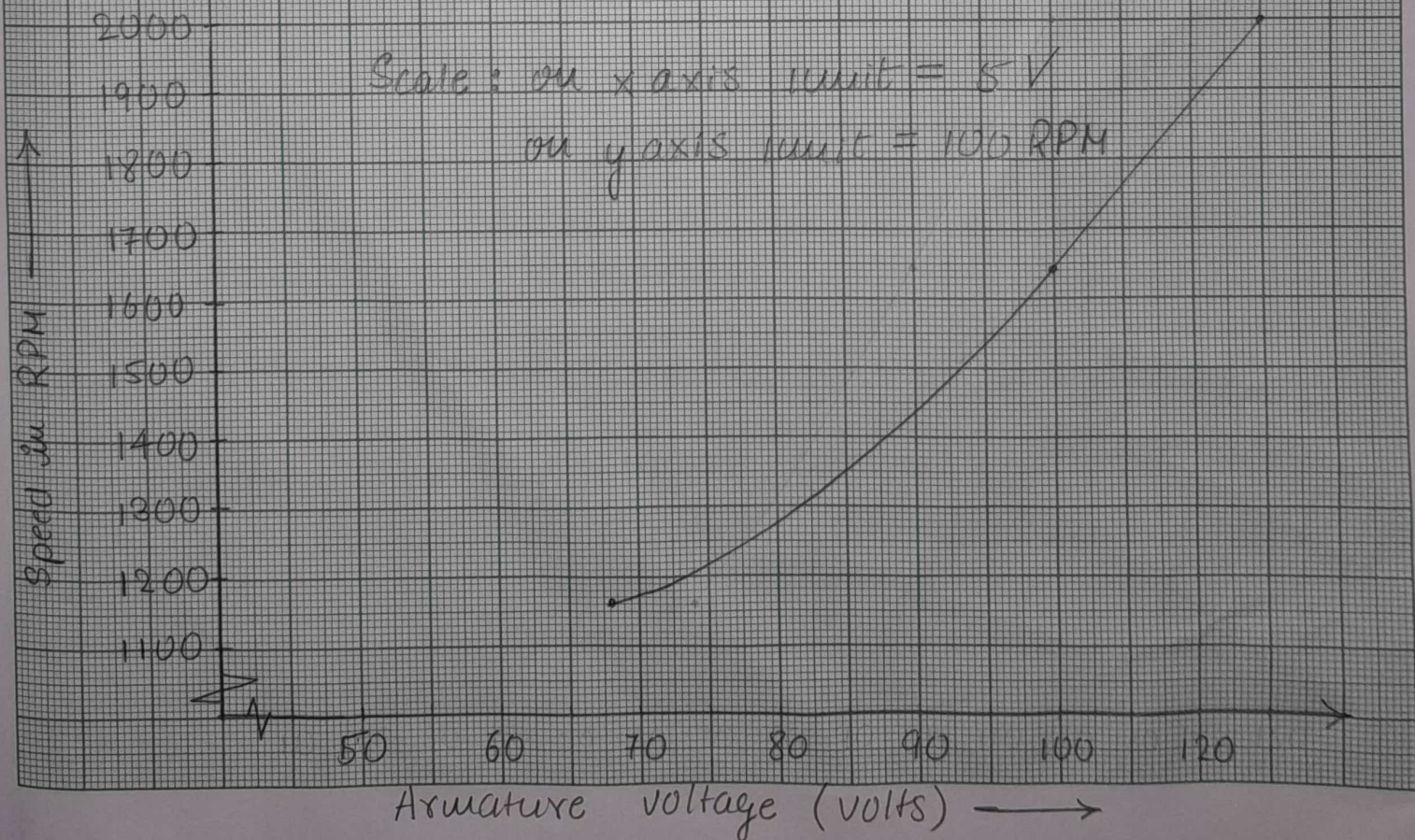
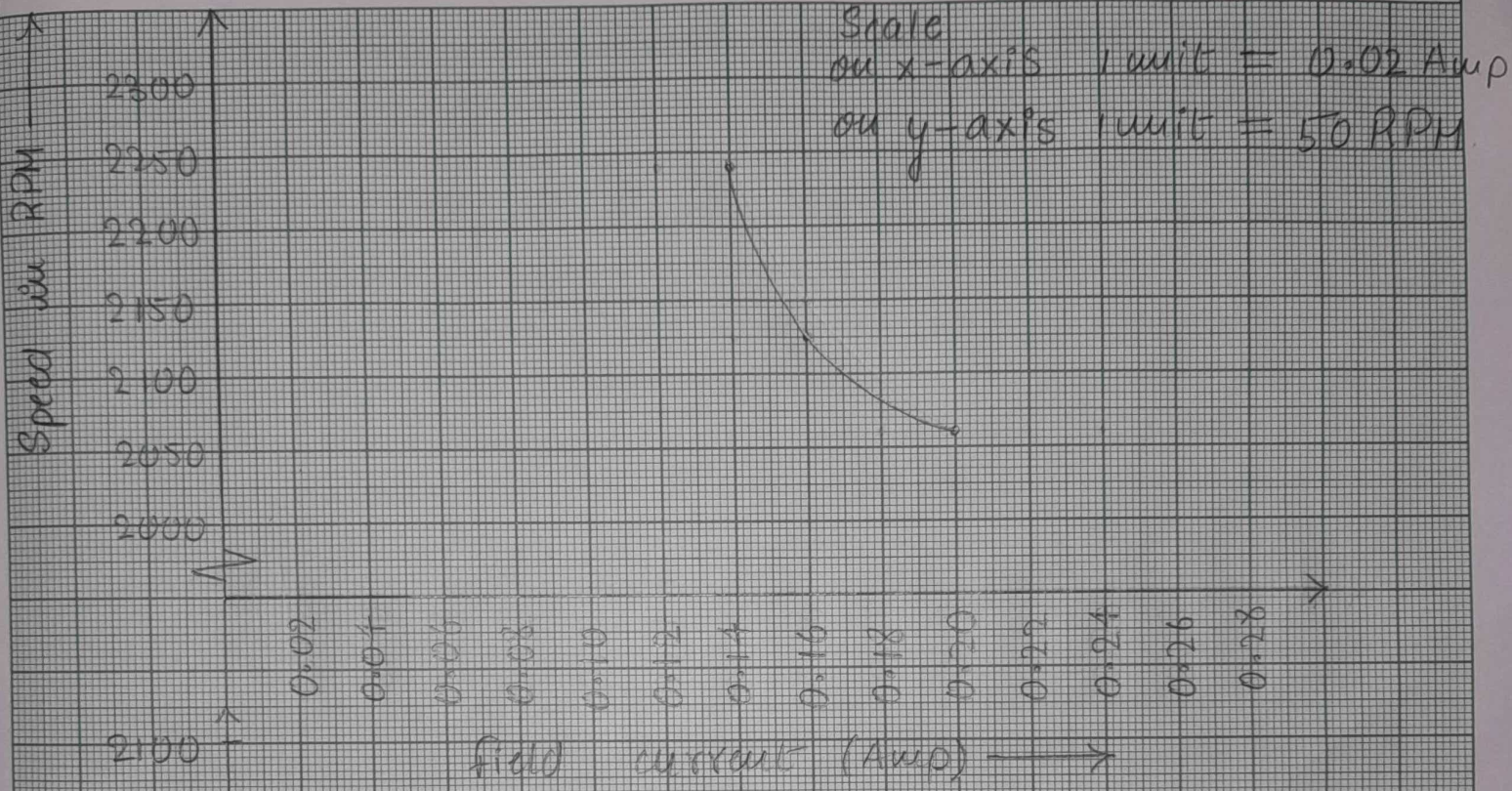
Electric drives are very commonly used for industrial and domestic applications. The reason for their wide usage is the ease with which their speed can be controlled and even reversed. The applications of electric motors range from fans, mixers and tape recorders, for domestic use to lifts, machine tool drives, rolling, and winding machines etc. for industrial purposes. In view of these applications some form of speed control is desired and the various methods of speed control used are the induction motor and the DC motor.

The DC motors are in general with more adaptable speed drives than AC motors which are associated with a constant-speed rotating field. Indeed one of the primary reasons for the strong competitive position of DC motors

Teacher's Signature

OBSERVATIONS:

Flux Control Method $V_a =$ volts (constant)			Armature Control Method $I_f =$ amp (constant)		
S. No.	Field current in Amp	Speed in RPM	S. No.	Armature voltage in Volts	Speed in RPM
1.	0.2	2060	1.	120	2000
2.	0.16	2126	2.	100	1640
3.	0.14	2240	3.	68	1160



in modern industrial drives is the wide range of speeds afforded.

The back emf in a DC motor:

$$E_b = \frac{(\phi Z N)}{60} \times \frac{P}{A}$$

$$E_b = K_a \phi \omega \text{ (Volts)}$$

where, E_b = back emf in volts

ϕ = Flux per pole in webers

Z = number of armature conductors

N = speed of motor in rpm.

P = No. of poles

ω = speed (rad/s)

A = number of parallel paths in the armature winding

$$E_b = V - I_a R_a$$

where, I_a = armature current

R_a = armature resistance

V = applied voltage

$$N = \frac{K_a (V - I_a R_a)}{\phi}$$

Since, the armature drop is small, it can be neglected

$$N = K_a (V/\phi)$$

This gives two methods of effecting speed changes i.e., the variation of field excitation, I and that of terminal voltage. The first method causes a change in the flux per pole, ϕ and is known as the field control and the second method is known as the armature control.

The field excitation can be varied by inserting a variable resistance in series with the field winding of a shunt motor. An increase in this resistance reduces the field flux and hence the speed increases above the normal speed of the motor (flux control method).

By inserting resistance in series with the armature, the E_b and hence, the speed of the motor drops below the normal speed (armature control method)

PROCEDURE:

- 1) Make the connections as per the circuit diagram

- 2) Set the resistance in field circuit (R_{sh}) and armature circuit (R_a) to zero and start the motor through starter with the rated voltage.

Flux control method :

- 1) Record the armature voltage, field current and the speed of the motor (N)
- 2) Increase the resistance R_{sh} in the field circuit in steps and note down the corresponding values of field current and the speed.

Armature control method :

- 1) Reduce the external resistance connected in the field circuit R_a to zero.
- 2) Note down the values of field current, armature voltage and the speed.
- 3) Increase the resistance R_a connected in the armature circuit in steps and record voltage across armature speed.

RESULT :